

18 – Core Data Model

Logical data model for the platform foundations: entity, location and tenancy scoping; stock keeping unit (SKU), unit of measure (UOM), batch and price ceilings; the immutable document base type with numbering series and compensating corrections; and the append-only audit log.

Field	Value
Document	18 – Core Data Model (design artefact; first output of the design phase, Stage 1)
Version	v0.2 – draft for design review (applies change requests CR-21-1, CR-21-2, sku_image, attributes, analyst role; aligns module ownership to doc 17 v0.2.1; adds Part E kernel event and sync tables)
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Status	Draft. Not yet reconciled against 17 – Solution Architecture module boundaries (see Section 8).
Inputs	02 Domain Model, 03 Document State Machines, 05 SRS, 06 NFR Specification, 07 Matrices, 09 Decisions Log (ADR-03, 07, 08, 10, 11, 14, 15, 18, 26, 27), 16 Questionnaire Answers, and the September 2026 review decisions (hosting neutrality, GRN per-shop series, multi-MRP resolution policy and gazetted control-price table, single-device shops).
Audience	Engineering leads and database designers. Business readers need only Sections 1, 2 and 9.
Scope boundary	Logical model at table and column level, with invariants. Physical tuning (indexes beyond the stated ones, partitioning sizes, storage parameters) belongs to the database design that follows.

How to read this document. Every table lists its columns, their types and the rules that bind them. Rules that span more than one table are collected as numbered invariants at the end of each part; those invariants are the contract that application code, database constraints and tests must all honour. Abbreviations are expanded on first use and collected in Section 10.

1. Modelling principles

Ten rules govern every table in this model. They are stated once here and applied throughout; where a table departs from one of them, the departure is called out explicitly.

#	Rule	Statement
P1	One owning entity per row	Every operational row carries <code>owner_entity_id</code> . A document between two legal entities carries both parties, but exactly one of them is the owner (the issuer). Tenancy scoping, visibility and audit all key off this column; there is no second mechanism.
P2	Identifiers are generated where the record is born	Primary keys are UUID version 7 (time-ordered universally unique identifiers). A till creates receipts offline with no central coordination; a UUIDv7 key is globally unique, sortable by creation time, and never collides on synchronisation. Business-facing numbers (document numbers, SKU codes) are separate attributes.
P3	Documents are immutable once issued	After issuance a document header and its lines are never updated or deleted. Any change is a new document that references the original (Part C). This is what makes the audit trail and the gapless series trustworthy.
P4	Ledgers are append-only	Stock movements, document state history and the audit log are insert-only tables. Balances are derived from movements, never edited directly. Database privileges enforce this: the application role holds INSERT only.
P5	Reference data is effective-dated	Prices, tax rates, control prices and conversion factors carry <code>effective_from / effective_to</code> . A past transaction can always be shown to have been correct at the time.
P6	One writer per record type	The till is the only writer of its own sale events and receipts; central is the only writer of catalogue, prices and rules, pushed to tills read-only. No table has two writers, so no merge logic exists (ADR-10).
P7	Trilingual text with English as the anchor	Names carry three columns: <code>name_en</code> (mandatory), <code>name_si</code> , <code>name_ta</code> . English is the universal fallback and the collation anchor; Sinhala and Tamil columns use ICU (International Components for Unicode) collations for correct sorting and search.
P8	Money and quantity are exact	Money is <code>numeric(14, 2)</code> in Sri Lankan rupees (LKR); quantities are <code>numeric(14, 3)</code> in the SKU base unit. No floating point anywhere.
P9	Time is stored twice	Every event stores <code>occurred_at</code> in UTC (Coordinated Universal Time) and <code>occurred_local</code> as the device clock reading. Offline devices have unreliable clocks; the server stamps <code>received_at</code> on arrival, and sequence numbers, not clocks, order events.
P10	Nothing is hard-deleted	Reference rows carry a <code>status</code> ; users are deactivated; documents are voided by a compensating document. The only DELETE privilege in the system belongs to migration tooling, not the application.

2. Model overview

The model has four parts, each owned by one platform module in the solution architecture. Arrows describe the direction of foreign-key dependency: nothing in Part A depends on later parts, so Part A can be built and tested first.

Part	Owning module	Tables	Depends on
A – Entity, location, tenancy	M1 Party, Tenancy & Security	<code>entity</code> , <code>entity_relationship</code> , <code>location</code> , <code>till_position</code> , <code>device</code> , <code>app_user</code> , <code>role</code> , <code>permission</code> , <code>role_permission</code> , <code>user_role</code> , <code>sod_pair</code>	–
B – Catalogue, batch, ceilings	M2 Catalogue & Batch (<code>uom</code> , <code>sku</code> , <code>conversions</code> , <code>barcodes</code> , <code>alias</code> , <code>image</code> , <code>tags</code> , <code>tax</code> , <code>batch</code>); M3 Pricing & Rules (<code>control_price</code> , <code>price_list</code> , <code>price_list_line</code> , <code>discount_rule</code>); M5 Inventory (<code>stock_lot</code> , <code>stock_movement</code> , <code>repack_recipe</code>)	<code>uom</code> , <code>sku</code> , <code>sku_uom_conversion</code> , <code>sku_barcode</code> , <code>sku_alias</code> , <code>sku_image</code> , <code>tag</code> , <code>sku_tag</code> , <code>tax_category</code> , <code>tax_rate</code> , <code>batch</code> , <code>stock_lot</code> , <code>stock_movement</code> , <code>control_price</code> , <code>price_list</code> , <code>price_list_line</code> , <code>discount_rule</code> , <code>repack_recipe</code>	A
C – Document base type	Kernel: Documents & Numbering base (document types owned by M4, M5, M6)	<code>document_type</code> , <code>numbering_series</code> , <code>document</code> , <code>document_line</code> , <code>document_link</code> , <code>document_state_history</code> , <code>document_attachment</code>	A, B
D – Audit log	Kernel: Audit	<code>audit_event</code> , <code>audit_event_type</code>	A (and references any table by type/id)

Part	Owning module	Tables	Depends on
E – Event backbone and sync state	Kernel: Event backbone; Edge sync gateway	event_outbox, event_inbox, device_sync_cursor, sync_quarantine, snapshot_change_log, idempotency_key	A

Module-specific tables (M4 extension tables, M7 customers and accounts, M8 projections) are introduced by their module designs and are not part of this document; a module design raises a change request here only when a shared table changes. Part E holds the kernel tables that docs 19 and 32 rely on; the till's local copies of parts C–E follow the same logical model (section 7).

3. Part A – Entity, location and tenancy scoping

The federation has three kinds of legal entity and one kind of place. A **shop is a location, not an entity**: it holds stock and issues receipts, but the legal party to every transaction is the multi-purpose cooperative society (MPCS) or distributor that owns it. This distinction is the single most important structural fact in the model and is enforced by the absence of any entity-level attributes on `location`.

Table `entity`

A legal party: the Federation, a distributor (itself a large MPCS with a distribution role) or an MPCS. Approximately 876 rows at go-live.

Column	Type	Meaning and rules
<code>entity_id</code>	uuid PK	UUIDv7.
<code>entity_code</code>	varchar(12) UNIQUE	Human-facing code used as a prefix in document numbers, e.g. M042. Assigned centrally, never changed.
<code>entity_type</code>	enum	FEDERATION DISTRIBUTOR MPCS. Exactly one FEDERATION row exists. A distributor is a legal MPCS carrying the distribution role; the enum records the role it plays in trading tiers.
<code>legal_name_en / _si / _ta</code>	text	Registered name; English mandatory (P7).
<code>registration_no</code>	varchar(40)	Co-operative registration number.
<code>vat_registration_no</code>	varchar(20)	VAT (Value Added Tax) registration. Required for all chain entities (J-03); printed on tax invoices of both parties.
<code>district</code>	varchar(40)	One of the 25 districts. Used for rollout waves and regional reporting.
<code>financial_year_start_month</code>	smallint	Common across the federation (J-08) but stored per entity so that a divergence is a data change, not a schema change.
<code>default_language</code>	enum	en si ta. Fallback for locations that do not set their own.
<code>costing_policy</code>	enum	WEIGHTED_AVERAGE only in phase 1 (E-03). Entity-level, no per-transaction override.
<code>credit_exposure_warn_only</code>	boolean	Always true in phase 1 (ADR-12: warn, never block). Present so the policy is data, not code.
<code>responsible_officer_user_id</code>	uuid FK app_user (nullable)	CR-21-1: the entity's named responsible officer under the PDPA and the data-governance agreement (FR-GOV-060); required for activation; changes audited.
<code>data_governance_signed_on</code>	date (nullable)	CR-21-1: date the entity signed the data-governance agreement (M-11).
<code>status</code>	enum	ONBOARDING ACTIVE SUSPENDED. Societies cannot leave (M-07); there is no CLOSED state.
<code>created_at, updated_at</code>	timestampz	Row bookkeeping.

- The Federation row has `entity_type = FEDERATION` and is the only entity whose users may hold the federation-wide read-only view (Section 3.7).
- There is no parent/child column between entities. Membership of the federation is implied by existence; trading tiers are expressed by `entity_relationship`, not by hierarchy.

Table `entity_relationship`

One row per trading pair. This is where commercial terms live: the price list, credit limit, payment terms and dispute time limit that A-06/A-07/A-09/D-06 say are agreed bilaterally.

Column	Type	Meaning and rules
<code>relationship_id</code>	uuid PK	
<code>seller_entity_id</code>	uuid FK entity	The supplying party.
<code>buyer_entity_id</code>	uuid FK entity	The receiving party. <code>seller <> buyer</code> enforced.
<code>price_list_id</code>	uuid FK price_list	The effective-dated list governing this pair (Part B). Federation-to-distributor pairs all point at the one uniform list; distributor-to-MPCS pairs each have their own.
<code>credit_limit</code>	numeric(14,2)	Informative exposure ceiling (ADR-12). Changes are permission-gated and audited (A-11).

Column	Type	Meaning and rules
payment_terms_days	smallint	Credit period used for ageing buckets.
discrepancy_window_days	smallint	Time limit within which a receipt discrepancy must be raised (D-06).
order_lock_hours_before_eta	smallint	Seller policy for the amend/cancel lock (C-08); per-order override recorded on the order.
allocation_rule	enum	FCFS default (C-03); configurable per seller, audited on change (C-04).
status	enum	ACTIVE SUSPENDED.
effective_from, effective_to	date	Relationships are effective-dated so that a re-negotiated price list is a new row, preserving history (P5).

- Uniqueness: one ACTIVE row per (seller, buyer) at any date. Overlapping effective ranges are rejected by an exclusion constraint.
- Intra-entity movements (Federation warehouse to warehouse; MPCS shop to sibling shop) are **not** relationships. They are transfer documents inside one entity, at cost, with no invoice (C-10/C-11, E-01).

Table location

A physical place belonging to exactly one entity: a Federation warehouse, a distributor depot, an MPCS shop or an office. A shop is one stock location with no shelf/back-store split (E-02).

Column	Type	Meaning and rules
location_id	uuid PK	
owner_entity_id	uuid FK entity	The legal owner (P1). Never changes; a shop transferred between societies is a new location with stock moved by document.
location_code	varchar(12)	Unique within the owner entity; used in shop-scoped document series, e.g. M042-S07.
location_type	enum	WAREHOUSE SHOP OFFICE. Only WAREHOUSE and SHOP hold stock.
name_en / _si / _ta	text	Display name.
address, district, geo_lat, geo_lng	text / numeric	For delivery notes with multiple drop points (D-02) and rollout planning.
language	enum	Per-shop interface and receipt language (P-02, H-09). Null means inherit from entity.
trading_hours	jsonb	Weekly pattern; used to schedule automatic updates outside trading hours (ADR-24).
size_band	enum	S M L planning attribute (H-06 open data item). Drives default till count and hardware profile.
connectivity_spec_met	boolean	Go-live gate flag per site (Q-06).
primary_till_position_id	uuid FK till_position	The till that holds the location-scoped numbering series (Part C, GRN decision). Reassignment is an audited event.
status	enum	PLANNED ONBOARDING ACTIVE DORMANT.

- Stock is held at location level. stock_lot (Part B) references location, never entity directly; entity ownership is derived through the location.

Table till_position

A logical checkout lane at a shop. Document series for till-issued documents bind to the position, not to hardware, so a replaced terminal continues the same series (ADR-11).

Column	Type	Meaning and rules
till_position_id	uuid PK	
location_id	uuid FK location	Must be a SHOP.
position_no	smallint	1..n within the shop; unique per location. Appears in receipt numbers.
status	enum	ACTIVE RETIRED. Retired positions keep their series history; numbers are never reused.

Table device

A physical POS terminal or back-office workstation. Devices are assigned to positions; the assignment is the only link between hardware and numbering.

Column	Type	Meaning and rules
device_id	uuid PK	
hardware_serial	varchar(64) UNIQUE	Manufacturer serial; the enrolment identity.
device_kind	enum	POS_TERMINAL WORKSTATION DRIVER_MOBILE.
owner_entity_id	uuid FK entity	P1.
current_till_position_id	uuid FK till_position	Null for non-POS devices. Assignment history lives in the audit log, not here.
enrolled_at, last_seen_at	timestampz	Fleet health; feeds the data-freshness stamp on federation reports (O-03).
app_version	varchar(20)	For the minimum-version floor enforcement (ADR-24).
status	enum	ENROLLED ACTIVE SUSPENDED RETIRED.

Users, roles and permissions

Role-based access control (RBAC) is dynamic: a permission catalogue, a role builder, and default templates (FR-SEC-010). The model keeps permissions as data so that entities shape their own roles without a release.

Table	Key columns	Rules
app_user	user_id PK; home_entity_id FK; username; pin_hash (till); password_hash (back office); mfa_secret (nullable); display_name; language; status	One user belongs to one home entity. Till operators authenticate by short numeric PIN (N-03); back-office by password, with TOTP (time-based one-time password) MFA (multi-factor authentication) required for financial and administrative functions (N-04). Deactivated, never deleted (N-08).
permission	permission_code PK; module; description_en; offline_allowed boolean; requires_mfa boolean	The catalogue. offline_allowed is what the till caches (H-02); requires_mfa marks online-only financial/admin actions.
role	role_id PK; owner_entity_id FK (null = federation template); name_en/_si/_ta; is_template	Templates are owned by no entity and cloned on adoption. The fourteen defaults in doc 07 plus the new shop-in-charge template (single-staff shops) ship as templates.
role_permission	role_id FK; permission_code FK	Many-to-many.
user_role	user_id FK; role_id FK; scope_entity_id FK; scope_location_id FK (nullable)	The scope columns implement M-08: an MPCS administrator holds an admin role scoped to its own entity and may grant roles scoped to its shops; the Federation holds no downward administrative scope.

Table sod_pair

CR-21-2. Segregation-of-duties pairs as data, with their enforcement mode (doc 19 section 3.2).

Column	Type	Meaning and rules
sod_pair_id	uuid PK	
permission_a, permission_b	FK permission	The conflicting pair, unordered.
mode	enum	INSTANCE (same user may hold both; may not act as both on one document) ROLE (may not coexist in one role).
owner_entity_id	uuid FK entity (nullable)	Null for the federation defaults (writeoff.request/approve, writeoff.request/witness, adjust.request/approve, user.manage/audit.review-self); an entity row overrides the default for that entity.
changed_by, changed_at		Audited (sod_pair.changed.v1).

3.7 Tenancy scoping and visibility enforcement

The system is one shared database. Ownership is a property of rows, not of schemas (M section preface, doc 15). Visibility is enforced at three layers, and all three key off the same two facts: the caller's scope (entity and optional location from user_role) and the row's owner or parties.

Layer	Mechanism	What it guarantees
Session scope	Every database connection sets <code>app.scope_entity_id</code> , <code>app.scope_location_id</code> and <code>app.scope_class</code> from the authenticated user before any query.	No query runs without a declared scope; a missing scope fails closed.
Row-level security (RLS) policy classes	Four policy classes applied to every operational table: OWN (rows where <code>owner_entity_id = scope</code>); PARTY (documents where scope is seller or buyer – read only the trading-facing columns); FEDERATION_VIEW (SELECT on everything, no writes, granted only to Federation roles – M-02 to M-04); EXTERNAL_TIMEBOXED (regulator and auditor roles, SELECT only, with <code>valid_until</code> , every grant audited – M-10, J-09).	A distributor cannot see an MPCS's internal data beyond the documents they trade (FR-GOV-020); a shop sees nothing of a sibling shop (M-05) because shop-scoped users carry a location scope and shop-internal tables filter on it.
Column masking views	For PARTY access, trading documents are exposed through views that omit the counterparty's internal columns (cost, margin, internal notes).	The buyer sees the seller's invoice, not the seller's cost basis.

Invariants for Part A.

- A-I1. Every operational table has a non-null `owner_entity_id` or derives one through a non-null `location_id`.
- A-I2. `location.owner_entity_id` is immutable after creation.
- A-I3. At most one ACTIVE `entity_relationship` per (seller, buyer) at any date.
- A-I4. A `till_position` belongs to exactly one SHOP location and its `position_no` is unique within it; positions are retired, never deleted.
- A-I5. A device holds at most one till position at a time; the assignment change is an audited event carrying old and new position.
- A-I6. No user may hold a role whose scope entity differs from their home entity unless the role class is `FEDERATION_VIEW` or `EXTERNAL_TIMEBOXED`.
- A-I7. Every location of type SHOP has a `primary_till_position_id` before it can issue any location-scoped document.

4. Part B – Catalogue, units, batches and price ceilings

Three ideas structure this part. The **SKU** is what is counted and sold; the **batch** is a physical supply identity carrying expiry and the printed maximum retail price (MRP); the **stock lot** is an entity's holding of a batch at a location, carrying that entity's cost. Separating batch from lot is what allows one physical batch to be owned successively by the Federation, a distributor and an MPCS at three different costs.

Table uom

Units of measure. A small, federation-governed vocabulary.

Column	Type	Meaning and rules
uom_code	varchar(10) PK	EA, KG, G, L, ML, CASE, PKT, BAG and similar.
name_en / _si / _ta	text	
is_weight	boolean	Weight units drive the weigh-station and per-kg pricing path (B-06, B-11).

Table sku

The smallest distinctly tracked and counted item, in one global namespace (B-01). Codes are system-generated and meaningless; the English short name is the mandatory universal field.

Column	Type	Meaning and rules
sku_id	uuid PK	
sku_code	varchar(12) UNIQUE	System-generated, opaque, never reused. Printed on shelf labels and used for till code lookup (B-11).
owner_entity_id	uuid FK entity	The steward. Federation for shared SKUs; the creating MPCS or distributor for local ones (B-08).
status	enum	DRAFT LOCAL SHARED INACTIVE. Promotion LOCAL→SHARED goes through sku_alias (B-10). INACTIVE while stock remains is parked (B-17) but the state exists.
short_name_en	varchar(80) NOT NULL	Mandatory (P7). short_name_si, short_name_ta optional for local SKUs, mandatory for SHARED.
description_en / _si / _ta	text	
base_uom_code	FK uom	Unit in which stock balances are held. Weighed goods use KG.
sold_by_weight	boolean	Weigh-and-price path at the till.
batch_tracked	boolean	Whether receipts must capture a batch (B-04). If false, a synthetic per-GRN batch is still created so that MRP and cost always have a home – see batch.is_synthetic.
expiry_tracked	boolean	Subset of batch-tracked categories; enables FEFO (first expired, first out) and expiry warning windows (F-06).
expiry_warning_days	smallint	Category default copied at creation; editable per SKU.
tax_category_id	uuid FK tax_category	Per-SKU tax category (G-09).
multi_mrp_policy	enum	AUTO_LOWEST (default) BARCODE_RESOLVED PICKER. The resolution policy adopted from the September 2026 study; overridable by tag. See 4.9.
origin_kind	enum	PURCHASED REPACK_OUTPUT LOCAL_SOURCED.
attributes	jsonb	Bounded, entity-owned free attributes (doc 20 Appendix A): non-searchable, no behaviour, validated against a size limit only. The escape hatch for local fields without schema change.
created_by, created_at		Creation is RBAC-controlled (B-07).

Table sku_uom_conversion

Pack conversions for the same physical item (B-02, ADR-07): a case of 24 is the same SKU with factor 24. A genuinely different retail item is a second SKU produced by a repack recipe, not a conversion.

Column	Type	Meaning and rules
sku_id, uom_code	PK (composite)	

Column	Type	Meaning and rules
factor_to_base	numeric(14,6)	One uom_code = factor × base units. Defined per SKU by the owner (B-13).
effective_from, effective_to	date	P5; a supplier changing case size is a new row.

Table sku_barcode

Many barcodes to one SKU is native (B-12). The optional batch link is what makes barcode-resolved MRP possible: an in-store batch sticker or a manufacturer barcode change resolves the batch at scan time.

Column	Type	Meaning and rules
barcode	varchar(48)	The scanned string after the scan layer has stripped any GS1 envelope. Unique together with symbology.
symbology	enum	EAN13 EAN8 UPCA GS1_128 GS1_DATAMATRIX GS1_QR INTERNAL. INTERNAL covers weigh-station 20–29 prefix labels and batch stickers.
sku_id	uuid FK sku	
uom_code	FK uom	A case barcode maps to the CASE unit of the same SKU.
batch_id	uuid FK batch (nullable)	Set when this barcode identifies one batch (batch sticker, MRP-wise label, or a 2D code carrying application identifier (AI) 10). Null for ordinary factory barcodes.
owner_entity_id	uuid FK entity	Who registered it; MPCS-printed stickers are owned by the MPCS and not visible to others.
status	enum	ACTIVE RETIRED.

Table sku_image

Product images (September 2026 decision): one per SKU, optionally one per barcode; stored in object storage; tills cache thumbnails for their assortment only.

Column	Type	Meaning and rules
image_id	uuid PK	
sku_id	uuid FK sku	
barcode	varchar(48) (nullable)	Set when the image is specific to a pack artwork (usually coincides with a barcode change).
owner_entity_id	uuid FK entity	Federation for shared SKUs; the local owner otherwise.
object_key_full, object_key_thumb	text	Keys in object storage; the thumbnail (about 128 px, under 10 KB) is what rides in the till snapshot.
content_hash	char(64)	Integrity of the full image.
status	enum	ACTIVE RETIRED.
attached_by, attached_at		Audited.

Table sku_alias

Preserves an originating entity's sales history when a local SKU is promoted or merged into a shared one (B-10, FR-SKU-100).

Column	Type	Meaning and rules
alias_sku_id	uuid FK sku PK	The retired local SKU.
canonical_sku_id	uuid FK sku	The shared SKU it now represents.
effective_from	date	Reports resolve aliases at query time so trend lines are unbroken.

Tags, tax

Table	Key columns	Rules
tag`, `sku_tag	tag_code PK; name_en/si/ta; governed boolean; owner_entity_id	A governed vocabulary replaces a mandatory hierarchy (B-15). Tags carry policy: expiry windows, loss tolerances (F-10), variance thresholds (E-07) and multi-MRP policy overrides all attach to tags or SKUs, with SKU winning over tag.

Table	Key columns	Rules
tax_category`, `tax_rate	tax_category_id PK; code; rate_percent; effective_from/to	Standard VAT only (J-03). Rates are effective-dated so receipts are reproducible. Inclusive pricing is computed from the rate in force at issuance.

Table batch

A physical supply identity for a SKU, created at the point stock first enters the federation (Federation GRN against a supplier, or an MPCs receipt of locally sourced goods). Batch is global; it is not owned by the entity holding it.

Column	Type	Meaning and rules
batch_id	uuid PK	
sku_id	uuid FK sku	
batch_no	varchar(40)	Manufacturer/supplier batch number as printed; unique with sku_id and supplier_id.
supplier_id	uuid FK supplier	Batch-level supplier attribution preserves traceability whatever the answer to B-03.
manufacture_date, expiry_date	date	Expiry mandatory when sku.expiry_tracked.
printed_mrp	numeric(14,2)	The MRP printed on the pack (ADR-08). Two concurrent batches may legally differ. Null only for goods with no printed MRP.
origin_document_id	uuid FK document	The GRN or repack document that created the batch.
is_synthetic	boolean	True when the SKU is not batch-tracked and the batch exists only to carry MRP and cost per receipt.
created_at	timestampz	

Table stock_lot

An entity's holding of one batch at one location: the unit of costing and of on-hand balance. Balances are derived from stock_movement; this table is the materialised current position.

Column	Type	Meaning and rules
stock_lot_id	uuid PK	
location_id	uuid FK location	Owner entity derived (A-11).
batch_id	uuid FK batch	
qty_on_hand	numeric(14,3)	In base units. May be negative (E-09): every negative occurrence is an exception-report event.
unit_cost	numeric(14,4)	This entity's acquisition cost per base unit, from its GRN (firm sale at GRN, ADR-03) or repack recipe. Landed cost flows in here for imports (L-05, phase 2).
received_at	timestampz	Drives stock ageing (E-11).
last_movement_seq	bigint	Sequence of the last applied movement; guards idempotent re-application on sync replay.

- Weighted-average cost of goods sold is computed per entity across its lots of a SKU at issue time (E-03). The lot keeps the acquisition cost; the entity's running average is a derived figure, so both IAS 2 cost and batch-level traceability survive.

Table stock_movement

Append-only ledger of every quantity change. The only way stock changes.

Column	Type	Meaning and rules
movement_id	uuid PK	
movement_seq	bigint	Monotonic per location, assigned by the location's writer (the till or the back office); the tie-breaker for ordering offline events (P9).
location_id, batch_id	FK	
qty_delta	numeric(14,3)	Signed, base units.

Column	Type	Meaning and rules
movement_type	enum	RECEIPT SALE SALE_REVERSAL TRANSFER_OUT TRANSFER_IN REPACK_CONSUME REPACK_PRODUCE COUNT_ADJUST WRITE_OFF RETURN_TO_SELLER OPENING_BALANCE. Loss categories (F-01) are on the write-off document, not here.
unit_cost_at_movement	numeric(14,4)	Cost basis frozen at the time; allows cost-of-goods-sold reproduction.
document_id, document_line_id	uuid FK	Every movement cites its document. No orphan movements.
occurred_at, occurred_local, received_at	timestampz	P9.
operator_user_id, device_id	uuid FK	Attribution (H-13).

Table control_price

Government-gazetted maximum price. New in this model: a date-effective, SKU-level ceiling that binds all stock regardless of the MRP printed on older packs (Consumer Affairs Authority Act, s.20(5); September 2026 study). Kept separate from batch.printed_mrp because the two ceilings have different scopes and lifecycles.

Column	Type	Meaning and rules
control_price_id	uuid PK	
sku_id	uuid FK sku (nullable)	Exactly one of sku_id or tag_code is set. Tag scope covers gazettes that name a category (e.g. all rice grades).
tag_code	FK tag (nullable)	
ceiling_price	numeric(14,2)	Per ceiling_uom_code, tax inclusive.
ceiling_uom_code	FK uom	Gazettes are typically per kg or per pack; conversion uses sku_uom_conversion.
effective_from	date NOT NULL	Date of gazette publication (the order operates from that date).
effective_to	date	Null while in force; set when rescinded or superseded.
gazette_reference	varchar(80)	Gazette or direction number, for the receipt footer and for audit.
entered_by, entered_at		Federation catalogue steward only; every insert is an audited event. There is no update: a correction is a new row and an effective_to on the old one.

- Enforcement is a hard block, never a warning: no receipt line, price-list line, discount rule outcome or printed label may exceed the control price in force on the transaction date (G-03).
- Federation is the only writer (P6). Tills cache the table; a till whose cache is older than a configured age warns the operator that ceilings may be stale.

Price lists and rules (M3), and the repack recipe (M5); summarised here because Part C documents cite them

Table	Key columns	Rules
price_list	price_list_id PK; owner_entity_id; kind (TRADE RETAIL); name; status	TRADE lists price an entity_relationship; a RETAIL list is one MPC's single retail list for all its shops (G-01). Federation recommended prices are a RETAIL-kind list with advisory = true.
price_list_line	price_list_id FK; sku_id FK; uom_code; price; volume_tier_from_qty; effective_from/to	Effective-dated (P5). Volume tiers for distributor-to-MPCS lists (A-07). Authoring validation: price must not exceed the printed MRP of any batch currently in stock at the owner's locations, nor any control price in force (the three-point MRP rule, item 11 of the September review).
discount_rule	rule_id PK; owner_entity_id; rule_kind; predicate jsonb; benefit jsonb; valid_from/to; status	The sole mechanism for any price reduction at a till (ADR-14). Every receipt line that benefits from a rule stamps rule_id, which is how markdowns are measured separately from promotions (F-07).
repack_recipe	recipe_id PK; owner_entity_id; input_sku_id; input_qty; output_sku_id; output_qty; expected_loss_pct	Bulk-to-pack transformation (B-06, ADR-07). Variance is measured against expected yield, not booked as shrinkage.

4.9 Price resolution at the till (data-level statement)

For a scanned line, the effective selling price is: **min(list price, lowest printed MRP among batches with positive on-hand at the location, control price in force)**, then discount rules apply, then the control price is re-checked as a hard cap. When `multi_mrp_policy = BARCODE_RESOLVED` and the scanned barcode carries a `batch_id`, that batch's MRP replaces the “lowest” term. When the policy is `PICKER`, the operator's choice is recorded on the line with the alternatives presented. In every case the receipt line stores `batch_id`, `mrp_applied`, `control_price_applied` and `cap_reason`, so the legality of every line is reproducible from the row alone.

Invariants for Part B.

- B-I1. `sku.short_name_en` is non-null for every status; `_si` and `_ta` are non-null for `SHARED`.
- B-I2. A barcode string is unique per symbology across the whole catalogue; two SKUs can never claim the same factory barcode. `INTERNAL` barcodes are unique within their owner entity.
- B-I3. Every `stock_lot` references a batch of the same SKU as the receiving document line.
- B-I4. `stock_lot.qty_on_hand` equals the sum of `stock_movement.qty_delta` for that lot; a nightly reconciliation job asserts it and raises an exception event on drift.
- B-I5. No `stock_movement` row exists without a `document_id`.
- B-I6. At most one `control_price` in force per (sku or tag) per date; overlapping rows are rejected by an exclusion constraint.
- B-I7. A `price_list_line` or discount-rule benefit that would yield a price above a control price in force is rejected at authoring, not merely at sale.
- B-I8. A batch's `printed_mrp` is immutable; a mis-keyed MRP is corrected by an audited batch-correction event that creates a new batch and re-points the lot.

5. Part C – The document base type

Every trading and stock artefact in the system is a document: order, delivery note, goods received note (GRN), invoice, credit and debit note, payment receipt, till receipt, write-off, count, adjustment, repack, transfer, opening balance. They share one header table, one line table and one lifecycle discipline. Type-specific data lives in extension tables keyed by `document_id` (for example the delivery note's drop points, the GRN's discrepancy lines); this document specifies the shared base only.

Table `document_type`

Registry of document kinds and their numbering behaviour. Adding a document kind is a data change plus an extension table, not a schema change to the base.

Column	Type	Meaning and rules
<code>doc_type_code</code>	<code>varchar(8)</code> PK	ORD, DN, GRN, INV, CN, DN2 (debit note), PRC (payment receipt), RCT (till receipt), WOF, CNT, ADJ, RPK, XFR, OPB.
<code>name_en / _si / _ta</code>	<code>text</code>	Printed heading.
<code>series_scope</code>	<code>enum</code>	ENTITY LOCATION TILL_POSITION. Decides which numbering series a document draws from (5.4).
<code>issuer_role</code>	<code>enum</code>	BUYER SELLER HOLDER. Who owns the document: the buyer issues an order, the seller issues a delivery note and invoice, the stock holder issues a GRN, count or write-off.
<code>bilateral</code>	<code>boolean</code>	True when a counterparty entity is mandatory (ORD, DN, GRN, INV, CN, DN2, PRC). False for internal stock documents.
<code>fiscal</code>	<code>boolean</code>	True for tax-facing documents (INV, CN, DN2, RCT) whose numbering must satisfy the Inland Revenue Department (IRD) expectations; the tax advisor's answers bind these. GRN is not fiscal (September 2026 decision).
<code>offline_issuable</code>	<code>boolean</code>	RCT, GRN (draft and confirm), CNT and WOF-draft are offline-issuable; ORD, INV, CN, PRC are not (H-02/H-03).
<code>posting_map</code>	<code>jsonb</code>	The designed journal-posting mapping for future accounting integration (J-02). Data, not code.

Table `numbering_series`

One row per series. A series is owned by an entity, a location or a till position according to the document type's scope. Sequential and gapless within the series; numbers are final at issuance and never reused; voids keep their numbers (ADR-11).

Column	Type	Meaning and rules
<code>series_id</code>	<code>uuid</code> PK	
<code>doc_type_code</code>	FK <code>document_type</code>	
<code>owner_entity_id</code>	<code>uuid</code> FK entity	Always set (P1).
<code>location_id</code>	<code>uuid</code> FK location (nullable)	Set for LOCATION and TILL_POSITION scope.
<code>till_position_id</code>	<code>uuid</code> FK till_position (nullable)	Set for TILL_POSITION scope only.
<code>prefix</code>	<code>varchar(24)</code>	Rendered as {entity_code}-{location_code}-{position}-{type} as applicable, e.g. M042-S07-T1-RCT, M042-S07-GRN, M042-INV.
<code>next_number</code>	<code>bigint</code>	Central view of the counter. For offline-issued scopes this is the last <i>synchronised</i> value; the authoritative counter lives on the holder device (5.5).
<code>holder_device_id</code>	<code>uuid</code> FK device (nullable)	For LOCATION-scope series: the shop's primary till (A-I7). For TILL_POSITION scope: the device currently assigned to the position. Null for ENTITY scope (central issues).
<code>fy_reset</code>	<code>boolean</code>	Whether the counter resets at financial-year start. Pending tax advisor answer ; default false.
<code>status</code>	<code>enum</code>	ACTIVE CLOSED. A closed series is never reopened; retiring a till position closes its series.

- Pre-reserved number ranges were evaluated and rejected (J-05, ADR-11). Each series has one counter with one writer; that is the whole mechanism.

Table document

The shared header. Immutable after issued_at is set (P3).

Column	Type	Meaning and rules
document_id	uuid PK	UUIDv7, generated by the issuing device (P2). Present from draft.
doc_type_code	FK document_type	
series_id	uuid FK numbering_series	Null while draft.
doc_number	bigint	Assigned at issuance from the series counter; null while draft. Unique with series_id.
doc_number_display	varchar(48)	Prefix + zero-padded number, frozen at issuance, e.g. M042-S07-T1-RCT-0001234.
owner_entity_id	uuid FK entity	The issuer (P1).
counterparty_entity_id	uuid FK entity (nullable)	Mandatory when the type is bilateral.
location_id	uuid FK location (nullable)	The stock location or shop concerned.
till_position_id, device_id	uuid FK (nullable)	For till-issued documents.
status	enum	Type-specific set (doc 03). The base enforces only DRAFT before issuance and terminal states after.
issued_at, issued_local	timestampz	P9. Immutability begins here.
business_date	date	The trading day the document belongs to, per the location's day-close; not the clock date.
operator_user_id	uuid FK app_user	Who issued it (H-13).
currency	char(3)	LKR; foreign currency only on Federation import procurement (I-08, phase 2).
net_amount, tax_amount, gross_amount	numeric(14,2)	Header totals, computed from lines at issuance and frozen.
reference_document_id	uuid FK document (nullable)	The primary upstream document: a DN references its ORD; a GRN its DN; an INV its DN or GRN. Correction relationships use document_link, not this column.
content_hash	char(64)	SHA-256 over the canonical serialisation of header and lines, computed at issuance. Verified on sync receipt; the P2 hash chain extends it.
origin	enum	ONLINE OFFLINE. Offline-issued documents carry their device sequence for gap checks.
device_seq	bigint (nullable)	The device's own monotonic sequence at issuance; the sync gap check compares consecutive values per series.
notes	text	Free text, frozen at issuance.

Table document_line

Lines are immutable with their header. Line-level facts needed to reproduce a price or a stock effect are stored here, not looked up later.

Column	Type	Meaning and rules
document_line_id	uuid PK	
document_id	uuid FK document	
line_no	smallint	Unique within document; frozen order.
sku_id	uuid FK sku	
batch_id	uuid FK batch (nullable)	Mandatory on stock-effecting lines (RCT, GRN, WOF, ADJ, RPK, XFR); the price-resolution batch on a receipt (4.9).
uom_code, qty	FK uom, numeric(14,3)	As transacted; base-unit quantity derived through the conversion in force.
unit_price	numeric(14,4)	Tax inclusive for retail lines (G-09).

Column	Type	Meaning and rules
mrp_applied, control_price_applied	numeric(14,2)	The ceilings that bound this line at issuance.
cap_reason	enum (nullable)	MRP_LOWEST MRP_BARCODE MRP_PICKED CONTROL_PRICE NONE.
discount_rule_id	uuid FK discount_rule (nullable)	The sole source of any reduction (ADR-14).
discount_amount, tax_rate_percent, tax_amount, line_total	numeric	Frozen.
unit_cost_at_issue	numeric(14,4)	For stock-effecting lines; cost basis at the moment.
loss_category	enum (nullable)	On WOF lines only: the F-01 categories.
reference_line_id	uuid FK document_line (nullable)	Upstream line (DN line for a GRN line, receipt line for a refund line).

Table document_link

Typed relationships between documents. This is the compensating-correction pattern: issued documents are never changed, so every correction is a new document pointing back.

Column	Type	Meaning and rules
from_document_id	uuid FK document	The later (correcting) document.
to_document_id	uuid FK document	The original.
link_type	enum	REVERSES (full void: same lines negated, e.g. till void, GRN reversal); CREDITS (credit note against invoice; refund receipt against receipt); DEBITS; ADJUSTS (count adjustment against a count; correction receipt with the delta only); SETTLES (payment receipt against specific invoices – I-07, never oldest-first); DISPUTES (discrepancy or claim against a GRN or DN); SUPERSEDES (re-issued draft, drafts only).
amount	numeric(14,2) (nullable)	For SETTLES and CREDITS: the amount applied to that specific original, so partial settlement across several invoices is explicit.
created_at, created_by		

- Rules: a REVERSES link may exist at most once per original and only while the original is in a reversible state per doc 03 (repack reversal eligibility is one such state rule). A reversed document and its reversal both keep their numbers. The net position of any original is the original plus the sum of its linked corrections; reports never re-read a corrected original alone.

Table document_state_history

Append-only record of every state transition (P4). The current document . status is the last row here.

Column	Type	Meaning and rules
history_id	uuid PK	
document_id	uuid FK document	
from_status, to_status	text	
occurred_at, occurred_local, received_at	timestampz	P9.
actor_user_id, device_id	uuid FK	
reason_code, reason_text	text	Mandatory for VOID, REJECT and any transition doc 03 marks as requiring a reason.

Table document_attachment

Photographs and files attached to discrepancy, claim and write-off documents (D-08, F-05).

Column	Type	Meaning and rules
attachment_id	uuid PK	
document_id	uuid FK document	
object_key	text	Key in object storage; the row is the index, the blob lives outside the database.
content_hash	char(64)	Integrity.
captured_at, captured_by, device_id		Captured offline, uploaded on sync; the document may issue before the upload completes.

5.5 Issuance protocol

Issuance is the moment a document becomes immutable and takes a number. The protocol is the same online and offline; only the location of the counter differs.

Step	Action	Guarantee
1	Validate the draft: parties, lines, ceilings (4.9), permissions (including <code>offline_allowed</code> when offline).	No invalid document ever takes a number.
2	Take the next number from the series counter. Online: a single-row <code>UPDATE ... RETURNING</code> on <code>numbering_series</code> inside the issuance transaction. Offline: the holder device increments its locally persisted counter in the same local transaction as the document write; the counter file is fsync-ed before the document is acknowledged.	Gapless and monotonic per series; a crash between steps leaves either both or neither.
3	Freeze totals, compute <code>content_hash</code> , set <code>issued_at</code> , write the DRAFT→ISSUED state row and the first audit event.	One transaction; the hash binds header, lines and number together.
4	Apply stock movements for stock-affecting types.	Movement rows cite the document (B-I5).
5	On sync, central verifies the hash, checks <code>device_seq</code> continuity per series, and advances the central <code>next_number</code> view. A gap is a sync anomaly event, never silently absorbed.	Central never re-numbers; the device's number is final.

Series scope by document type (the decision from the September review applied):

Scope	Document types	Counter holder
TILL_POSITION	RCT (till receipt, including voids and same-session refunds)	The device assigned to the position.
LOCATION	GRN, CNT (count), WOF (write-off draft), XFR (intra-entity transfer at a shop)	The shop's primary till (<code>location.primary_till_position_id</code>). A shop with one till is the trivial case.
ENTITY	ORD (buyer's series), DN and INV (seller's series), CN, DN2, PRC, ADJ, RPK, OPB	Central; issued only online.

Invariants for Part C.

- C-I1. No column of document or `document_line` changes after `issued_at` is non-null, except `status`, which changes only through `document_state_history`. Enforced by a database trigger that raises on any other UPDATE, and by the absence of UPDATE privilege on lines.
- C-I2. (`series_id`, `doc_number`) is unique and, within a series, dense: for every issued number $n > 1$, number $n-1$ exists. Verified by the nightly gap check and by the sync anomaly check.
- C-I3. A document's series scope matches its type's `series_scope`: a GRN always draws from a LOCATION series of the receiving shop; an order always from the buyer's ENTITY series.
- C-I4. A bilateral document has a non-null counterparty and the pair (owner, counterparty) has an `entity_relationship` in force on `business_date`.
- C-I5. Every issued document has exactly one DRAFT→ISSUED row in `document_state_history` with `occurred_at` = `document.issued_at`.
- C-I6. A correction never changes the original: the original's rows are byte-identical before and after the correction; the correction is discoverable only through `document_link`.
- C-I7. `content_hash` recomputed from stored rows equals the stored value; verified on sync and by a sampled nightly job.

6. Part D – Append-only audit log

The audit log answers “who did what, to which record, when, from where, and why” for every event in the catalogue of doc 07 section 3. It is deliberately simpler than the tables it observes: one wide table, insert-only, partitioned by month, retained indefinitely (J-06), with reserved columns for the phase-2 hash chain (N-05).

Table audit_event_type

The event catalogue as data, so that new audited events can be added without a schema change and the exception-report reviewer mapping (N-06) can attach to each type.

Column	Type	Meaning and rules
event_type_code	varchar(40) PK	e.g. PRICE_OVERRIDE (legacy, unused), VOID, REFUND, WRITE_OFF_REQUEST, WRITE_OFF_APPROVE, STOCK_ADJUST, NEGATIVE_STOCK_SALE, DRAWER_OPEN_NO_SALE, ROLE_CHANGE, USER_DEACTIVATE, SERIES_REGISTER, SERIES_HOLDER_CHANGE, DEVICE_POSITION_CHANGE, SYNC_ANOMALY, REGULATOR_GRANT, CREDIT_LIMIT_CHANGE, CONTROL_PRICE_ENTER, CONTROL_PRICE_CAP_APPLIED, MRP_PICKER_USED, MODE_SWITCH_SHOP_MGMT, PHONE_CHANGE (customer).
severity	enum	INFO REVIEW ALERT. REVIEW and ALERT feed exception reports.
reviewer_role_template	FK role (nullable)	The named reviewer requirement of N-06.
offline_capturable	boolean	Whether the till may raise it offline; all till events are.

Table audit_event

One row per audited action. Insert-only.

Column	Type	Meaning and rules
audit_id	uuid PK	UUIDv7 (P2), generated at the source device.
event_type_code	FK audit_event_type	
occurred_at, occurred_local, received_at	timestampz	P9. Partition key is received_at (monotonic at central), not occurred_at.
owner_entity_id, location_id	uuid FK	Scope of the action (P1); RLS applies to audit reads exactly as to operational reads.
actor_user_id	uuid FK app_user	The authenticated operator. Null only for system-generated events, which then carry actor_system.
actor_system	varchar(40) (nullable)	e.g. SYNC, NIGHTLY_RECON, UPDATE_AGENT.
device_id, till_position_id	uuid FK (nullable)	Where.
subject_table, subject_id	varchar(40), uuid	What was acted on. Polymorphic by design; the audit log must be able to reference any table without a foreign key to each.
document_id	uuid FK (nullable)	Set when the subject is, or belongs to, a document; indexed for per-document timelines.
before_state, after_state	jsonb (nullable)	Minimal diff of the affected columns, not whole rows; for immutable subjects before_state is null.
reason_code, reason_text	text (nullable)	Mandatory where the event type says so (voids, refunds, write-offs, overrides).
witness_user_id	uuid FK (nullable)	For write-offs: the distinct witness (F-05). For single-staff shops the September decision applies: photographic evidence mandatory and the MPSC approver acts as remote witness; the model records whoever fulfilled the role.
device_seq	bigint	Source device's monotonic sequence; gap detection on sync.
correlation_id	uuid	Groups the events of one user action (a refund raises REFUND plus DRAWER_OPEN plus the state transition).
prev_hash, row_hash	char(64) (nullable)	Reserved for the phase-2 hash chain. Null in phase 1; when enabled, row_hash = SHA-256(prev_hash canonical row) per partition.

- Enforcement is structural, not procedural: the application database role holds INSERT on `audit_event` and nothing else; no trigger, function or job in the application schema may UPDATE or DELETE it; partitions are detached to cold storage, never dropped, under the retention policy pending J-06.
- Personal data: the audit log stores identifiers, not PII (personally identifiable information). A customer phone change records the customer id and the fact of change; the values live in the customer table under PDPA (Personal Data Protection Act) rules (ADR-27).

Relationship to the sync event stream. The till's outbox carries three kinds of record to central: issued documents with their lines and state rows, stock movements, and audit events. All three are append-only at both ends and are replayed idempotently by primary key. The audit log is therefore complete at central only after sync, which is why federation-level exception reports carry the per-entity data-freshness stamp (O-03).

Invariants for Part D.

- D-I1. No UPDATE or DELETE ever succeeds on `audit_event` from the application role; a test asserts the privilege set on every deployment.
- D-I2. Every document state transition has a corresponding audit event with the same `correlation_id`.
- D-I3. Every event type marked REVIEW or ALERT has a non-null reviewer role template before it can be raised in production.
- D-I4. (`device_id`, `device_seq`) is unique and dense per device; gaps raise SYNC_ANOMALY.
- D-I5. Audit reads obey the same RLS policy classes as operational reads; the Federation and auditors read view-only.

6A. Part E – Event backbone and sync state (kernel)

The kernel tables docs 19 and 32 rely on. All are append-only or single-writer bookkeeping; none holds business facts that are not also in parts A–D. The till carries local equivalents of event_outbox and idempotency state.

Table event_outbox

Transactional outbox and archive (doc 19 6.2, doc 17 ADR-30): the system of record for domain events, written in the same transaction as the business change; partitioned by month; retained indefinitely.

Column	Type	Meaning and rules
event_id	uuid PK	UUIDv7 (P2).
event_type	text	Versioned dotted type, e.g. document.issued.v1.
occurred_at, occurred_local, received_at	timestampz	P9; partition key is received_at.
source	text	central or the device id.
source_seq	bigint	Dense per source; unique with source.
owner_entity_id, location_id	uuid FK	Routing and RLS scope.
aggregate_type, aggregate_id	text, uuid	
correlation_id, causation_id, actor_user_id	uuid	
payload	jsonb	Small; identifiers and changed facts.
published_at	timestampz (nullable)	Set by the relay; null means not yet published to the broker.

- Insert-only for the application role; the relay holds UPDATE on published_at only.

Table event_inbox

Per-consumer idempotency record (doc 19 6.2): which event ids a consumer has applied.

Column	Type	Meaning and rules
consumer	text	Consumer name (module.handler).
event_id	uuid	PK with consumer.
applied_at	timestampz	
outcome	enum	APPLIED SKIPPED FAILED; FAILED rows carry the last error for the dead-letter view.

Table device_sync_cursor

The only server-side sync state per device (doc 32 section 2); row-locked during ingestion.

Column	Type	Meaning and rules
device_id	uuid PK FK device	
last_applied_seq	bigint	Highest device_seq durably applied.
last_batch_id	uuid	For idempotent answers to a resent batch.
last_ack_at	timestampz	
snapshot_version_reported	bigint	The version the device says it trades on.
app_version_reported	varchar(20)	
pending_instructions	jsonb	Revoke, floor notice, re-request-from-seq, series holder change; cleared when acknowledged.

Table sync_quarantine

Events that could not be applied (doc 32 S4): stored raw, never lost, never blocking.

Column	Type	Meaning and rules
quarantine_id	uuid PK	
device_id, device_seq, event_id		Identity of the offending event.
raw_payload	bytea	Exactly as received.
reason_code, reason_text	text	Schema failure, hash mismatch, duplicate id, sequence anomaly.
received_at, resolved_at, resolved_by		Resolution is an audited manual action; the row is never deleted.

Table snapshot_change_log

Per-location change log that produces snapshot deltas (doc 32 5.2); retained for the delta window, then a full snapshot serves older devices.

Column	Type	Meaning and rules
location_id	uuid FK location	
version	bigint	Monotonic per location; PK with location_id.
table_name, row_id	text, uuid	What changed.
op	enum	UPSERT TOMBSTONE.
apply_from	date (nullable)	Business date the change takes effect on tills (prices, rules, control prices).
urgent	boolean	Control-price entry, permission revoke: download on next heartbeat.
created_at	timestampz	

Table idempotency_key

Result store for mutating web commands (doc 14 section 1.3): the same key returns the same result from any instance.

Column	Type	Meaning and rules
idempotency_key	text PK	Client-supplied header value, scoped by user.
user_id	uuid FK app_user	
request_hash	char(64)	A reused key with a different request body is rejected.
response_status, response_body	smallint, jsonb	Stored result.
created_at	timestampz	Rows expire after 24 hours.

Invariants for Part E.

- E-I1. Every business transaction that changes a table in parts A–C inserts at least one event_outbox row in the same transaction.
- E-I2. (source, source_seq) is unique and dense per source; central raises SYNC_ANOMALY on a gap that is not explained by sync_quarantine.
- E-I3. device_sync_cursor.last_applied_seq never decreases except through the audited sequence-reset procedure (doc 32 section 8).
- E-I4. A location's snapshot_change_log versions are dense; a delta is the contiguous range from the device's version + 1.

7. Cross-cutting conventions

Concern	Convention
Naming	snake_case tables and columns; singular table names; _id for UUID keys, _code for human codes; booleans as is_ or a verb phrase; every table has created_at.

Concern	Convention
Keys	UUIDv7 everywhere (P2). Natural keys (codes, barcodes, numbers) are unique constraints, never primary keys, so a code can be corrected without re-keying.
Language columns	Three columns per text field (P7). Search uses ICU collations <code>si-x-icu</code> and <code>ta-x-icu</code> on the Sinhala and Tamil columns and a trigram index on all three for partial-match lookup at the till (P-04). OpenSearch analysis is phase 2 and reads from these columns.
Time	<code>timestampz</code> in UTC plus a local reading (P9). <code>business_date</code> is a separate date column set by the shop's day-close, so a sale at 00:30 before Z-report belongs to the previous trading day.
Money and quantity	<code>numeric(14, 2)</code> LKR; <code>numeric(14, 4)</code> for unit cost and unit price; <code>numeric(14, 3)</code> for quantity. Cash-tender rounding to the rupee is a receipt-level field, not a change to line values (G-10).
Soft delete and status	No DELETE (P10). Every reference table has a status enum with an explicit terminal value.
Partitioning	<code>document</code> , <code>document_line</code> , <code>stock_movement</code> , <code>document_state_history</code> and <code>audit_event</code> are range-partitioned by month on <code>received_at</code> . At the planning envelope (about 3 million receipts a day) this keeps hot partitions indexable and lets cold partitions move to cheaper storage without touching the schema.
Row-level security	Enabled on every operational table from the first migration, with the four policy classes of 3.7. A table without an RLS policy fails the schema test.
Extension tables	Type-specific document data lives in <code>doc_grn</code> , <code>doc_delivery_drop</code> , <code>doc_writeoff</code> and similar, keyed one-to-one on <code>document_id</code> , inheriting immutability from the base by the same trigger.
Till local database	The till holds the same logical model for the tables it writes (documents, lines, state history, movements, audit) and read-only caches of what central pushes (<code>sku</code> , <code>sku_barcode</code> , batch and lot for its location, price lists, discount rules, control prices, permissions). Same column names, same types; the sync contract is a subset, not a translation.

8. Decision requests and open points

These are the points this model could not settle from the baseline. Each names the decider and what changes if the answer differs from the assumption used here.

#	Point	Assumed here	Decider	If different
DR-1	Financial-year reset of numbering series	No reset (<code>fy_reset = false</code>)	Tax advisor	Series counter reset job; display format gains a year segment; gap check becomes per-year.
DR-2	GRN outside the fiscal gapless obligation	Yes (internal document)	Tax advisor	GRN moves to ENTITY scope online-only, which delays ownership transfer during outages (conflicts with ADR-03).
DR-3	Control price scope by tag as well as by SKU	Both supported	Federation catalogue steward	Tag scope dropped; every gazetted category must be enumerated per SKU by the steward.
DR-4	Batch identity global vs per receiving entity	Global batch; per-entity cost in <code>stock_lot</code>	Engineering lead	Per-entity batches would fragment traceability and duplicate MRP entry at every tier.
DR-5	Synthetic batches for non-batch-tracked SKUs	Yes, one per GRN line	Engineering lead	MRP and cost would need a home on the lot instead; price resolution 4.9 would branch on tracking flag.
DR-6	Alignment with 17 – Solution Architecture module ownership	Closed in v0.2: ownership follows doc 17 v0.2.1 (M1–M9 and kernel)	–	–
DR-7	Single-staff write-off witness rule	Remote MPCs approver as witness with mandatory photo	Federation admins (ratification batch)	<code>witness_user_id</code> becomes nullable with a stated reason code instead.
DR-8	Control-price operational owner	Federation catalogue steward	Federation	Without a named owner the ceiling table goes stale and the hard block enforces wrong ceilings.

9. Traceability to the requirements baseline

Where each part of the model earns its place. Requirement identifiers are those of 05 SRS and the decision log.

Model element	Requirements and decisions
entity, entity_relationship, location (shop as location)	FR-PTY-010 to 070; ADR-03 (ownership at GRN); A-06, A-07, A-09, A-10, C-03, C-04, C-08, D-06; M-07 (no leaving)
till_position, device, numbering_series holder	ADR-11; J-05; FR-BIL-050, FR-BIL-060; September 2026 GRN series decision
app_user, role, permission, user_role scope; RLS classes	FR-SEC-010 to 070; FR-GOV-010 to 060; M-02 to M-06, M-08, M-10; N-03, N-04, N-08; J-09; ADR-18, ADR-26
sku, sku_uom_conversion, sku_barcode, sku_alias, tag	FR-SKU-010 to 110; ADR-07; B-01, B-02, B-08 to B-15, B-17
batch, stock_lot, stock_movement	FR-INV-010 to 100; ADR-08 (MRP on batch); E-03 (weighted average); E-04 (FEFO); E-09 (negative stock); B-04; F-01
control_price; price resolution 4.9	G-03 (gazetted ceilings as hard control); FR-PRC-040 (to be split per the September review); FR-PRC-030, 060; ADR-14; September 2026 multi-MRP study
document, document_line, document_link, state history	FR-BIL-010 to 150; doc 03 lifecycles; ADR-11; I-07 (settle to specific invoices); D-05, D-07, D-08; F-03 to F-05; H-02/H-03 (offline issuable set)
audit_event, audit_event_type	FR-SEC-050 (immutable audit); N-05, N-06, N-07, N-09; doc 07 section 3 event catalogue; J-06 retention; ADR-27 minimal PII
Part E kernel tables	ADR-30 (outbox archive); doc 19 6.2; doc 32 sections 2, 3, 5, 7; doc 14 1.3 (idempotency)
sod_pair, entity officer fields, sku_image, sku.attributes	CR-21-1, CR-21-2 (doc 21); doc 20 Appendix A (images, attributes, analyst role – the analyst role is a permission row rpt .analyst .query in the M1 seed data, not a table)

10. Abbreviations used in this document

Term	Meaning
AI	Application Identifier – the numbered data fields inside a GS1 barcode (AI 01 product number, AI 10 batch, AI 17 expiry).
FEFO	First expired, first out – the picking rule for expiry-tracked goods.
GRN	Goods received note – the receiving entity's record of what actually arrived; the moment ownership transfers.
GS1	The global standards body for product identification and barcodes.
ICU	International Components for Unicode – the library providing correct sorting and comparison for Sinhala and Tamil.
IRD	Inland Revenue Department of Sri Lanka.
LKR	Sri Lankan rupee.
MFA / TOTP	Multi-factor authentication / time-based one-time password.
MPCS	Multi-purpose cooperative society – the legal entity that owns retail shops.
MRP	Maximum retail price printed on a pack.
PDPA	Personal Data Protection Act No. 9 of 2022 (Sri Lanka).
PII	Personally identifiable information.
POS	Point of sale – the till.
RBAC	Role-based access control.
RLS	Row-level security – database policies that filter rows by the caller's scope.
SKU	Stock keeping unit – the smallest distinctly tracked and counted item.
UOM	Unit of measure.
UTC	Coordinated Universal Time.
UUIDv7	Universally unique identifier, version 7 – time-ordered random identifier safe to generate on disconnected devices.
VAT	Value Added Tax.

End of document. Draft for design review; subject to revision after the decision requests in Section 8 are answered.